Project Report:

Network-integrated pre- and post-processing in scientific computing

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Contents

1 Context and Objectives **2**

2 Geometry and mesh generation **2**
2.0.1 Ring generation . . . 3
2.1 Helix generation . . . 3
2.2 Generation of the total geometry . . . 4
2.3 Mesh options . . . 5
2.4 Limitations . . . 5

3 Numerical simulation **6**
3.1 General overview . . . 6
3.2 Parallelization . . . 6

4 Visualization and analysis of results **7**
4.1 Hyperparameters . . . 7
4.2 Temperature Field Extraction and Heat Dissipation . . . 7
4.3 Current Density Distribution . . . 9
4.4 Electric Field Visualization . . . 11
4.5 Electric Potential Visualization . . . 12
4.6 Comparison Plots . . . 13
4.7 General Interpretation and Hypothesis . . . 14

5 Annexes **15**

1 Context and Objectives

This project was carried out as part of the "Network-integrated pre- and post-processing in scientific computing" module of the Master 2 CSMI and Master 1 CMI IIRVIJ programs.

The overall objective is to implement a digital pipeline from geometry generation to results analysis. The tools used are:

- Gmsh (Python API) for geometry and mesh generation (pre-processing)
- Feel++ and its thermoelectric toolbox for solving the physical problem
- Gaya computing cluster for running simulations
- Paraview for visualizing and analyzing numerical results (post-processing)

The study focuses on an electromagnet modeled as a thermoelectric insert, consisting of two conductive helices and a ring with cooling slits . This work aims to analyze the thermoelectric behavior of the object under imposed thermal and electrical conditions.

The project is structured in three parts:

- Generation of the geometry and mesh
- Setting up the problem and numerical solution
- Analysis and post-processing of results

2 Geometry and mesh generation

The objective of this first part is to generate the CAD geometry of the electromagnet and produce a 3D mesh that can be used by Feel++.

The generation was done entirely via the Gmsh Python API. Each component (helix and ring) is generated individually, then assembled into an overall geometry.

The generation of geometries is based on `.json` configuration files. These files describe all the geometric parameters of the elements, which are then used by the Python file. The parameters include:

- inner and outer radii
- axial terminals
- number of turns and pitch of the helices
- orientation of components
- additional geometric options (starting holes, slots, etc.)

2.0.1 Ring generation

Here are some illustrations obtained.

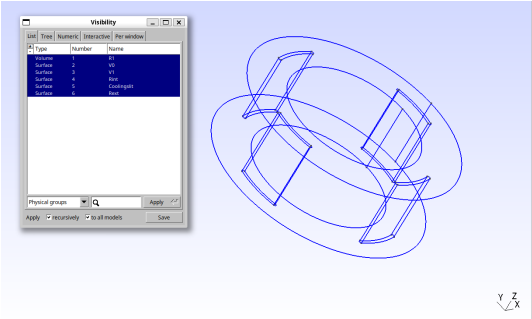


Figure 1: Ring geometry

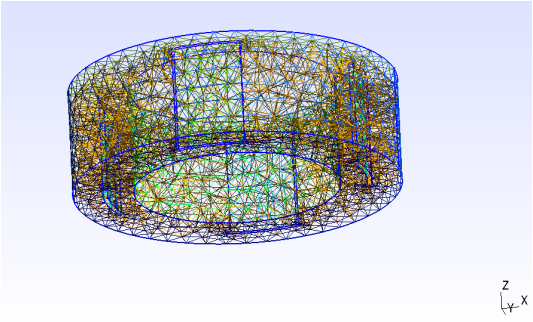


Figure 2: Ring mesh

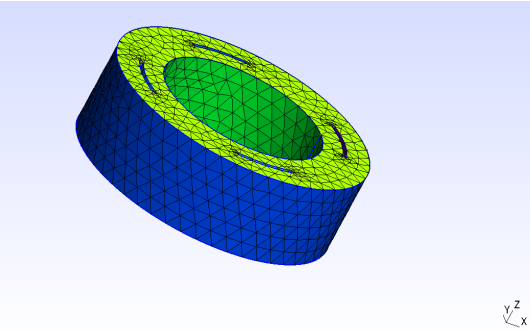


Figure 3: Surface mesh of the ring

2.1 Helix generation

Here are some examples obtained.

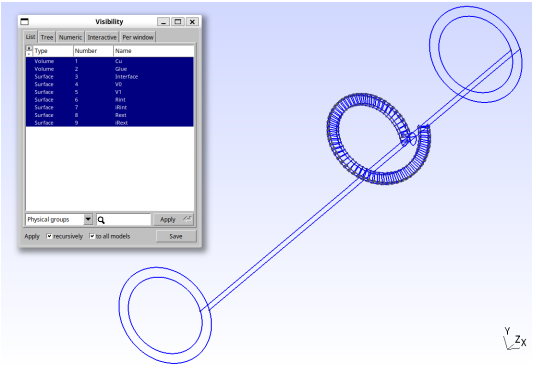


Figure 4: H1 helix geometry

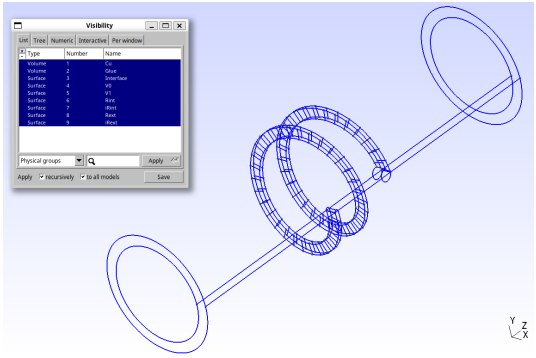


Figure 5: H2 helix geometry

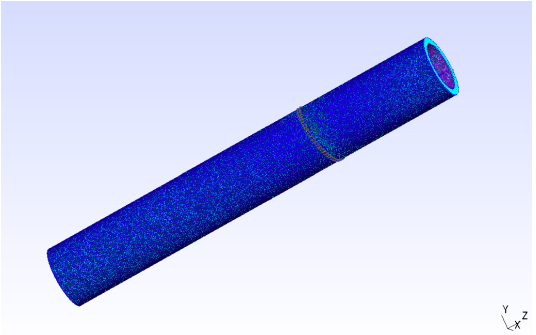


Figure 6: H1 helix mesh

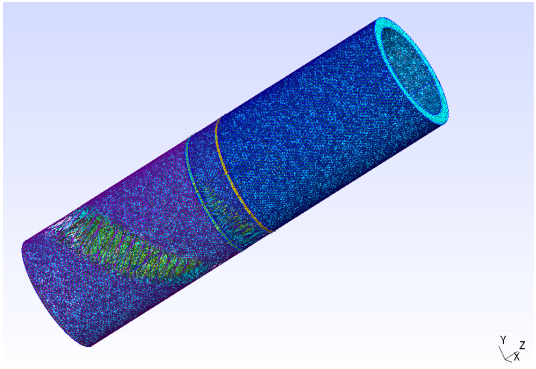


Figure 7: H1 helix mesh

2.2 Generation of the total geometry

The electromagnet assembly is then created using the `insert.py` script. Here is an example of the result:

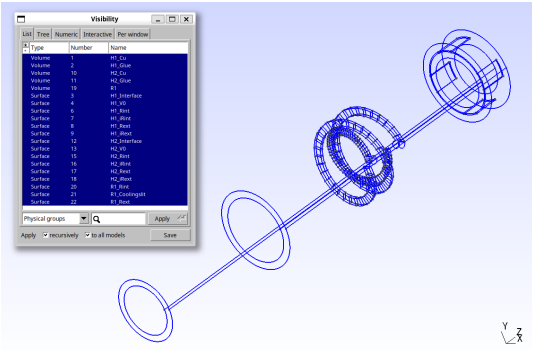


Figure 8: Insert geometry

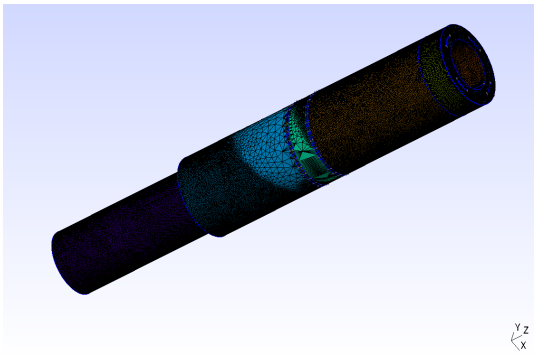


Figure 9: Insert mesh

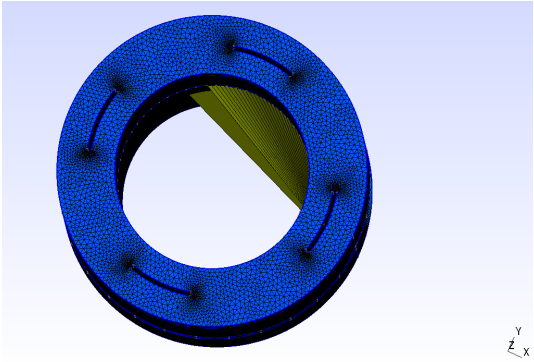


Figure 10: Top view, on the ring

2.3 Mesh options

Command line options have also been added to control mesh generation:

- `-mesh_dim` to select a 2D or 3D mesh
- `-algo2D` and `-algo3D` to explicitly choose the meshing algorithm (Delaunay, Frontal, or automatic)
- `-refine` to apply local refinement to the interfaces of the helices and the cooling slits of the ring. This increases the accuracy of these areas where strong thermal or electrical gradients are expected (but higher computation time).
- `-nopopup` to disable the Gmsh interface, which is essential when generating on gaya.

2.4 Limitations

Some 3D meshing algorithms were too restrictive for this geometry. The Delaunay algorithm, which is slightly faster, generated errors during volume generation. The Frontal Netgen algorithm was more robust (and was less restrictive), which only produced non-blocking warnings. However, Frontal Netgen is not available on gaya, so the mesh was generated locally before being transferred to the Gaya cluster for the simulation phase.

In any case, mesh generation cannot be parallelized on gaya and is done on a single computing core, so performing this step on gaya will not be any faster.

3 Numerical simulation

3.1 General overview

Thermoelectric simulations are performed on the Gaya cluster using the Feel++ thermoelectric toolbox. This toolbox solves a coupled problem of thermal conduction and electrical conduction.

Once the mesh has been validated, the simulation is based on three main files:

- a `.cfg` file defining the general configuration (discretization, solvers, output directories)
- a `.json` file describing the physical model (materials, boundary conditions, post-processing)
- a `.slurm` script for submitting the calculation to Gaya

The material properties are those of copper. The boundary conditions combine the imposed convective heat flux and electric potential. Several fields are exported for post-processing, such as temperature, electric potential, and current density. However, the latter is not used in this work.

3.2 Parallelization

Calculations can also be parallelized by partitioning the mesh into a number of subdomains equal to the number of cores used. Partitioning is performed using the Feel++ partitioning tool, although Gmsh also provides one.

In this project, the calculations performed are not extremely costly: a single simulation is performed on a given geometry, without in-depth parametric study (material variation, convergence study, or temporal resolution). Thus, parallelization remains optional but still significantly reduces computation time. For example, in the case studied, a simulation takes about 230 seconds on a single computing core and is reduced, after mesh partitioning, to about 64 seconds on 8 cores.

Here is an example illustrating mesh partitioning into 8 subdomains, on Paraview.

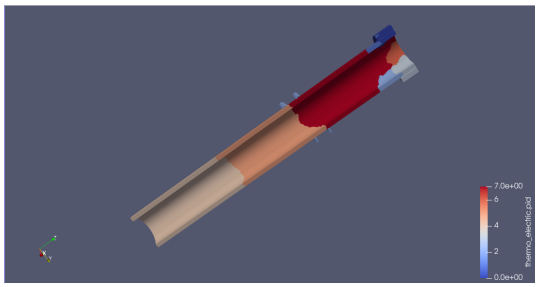


Figure 11: Partitioning into 8 subdomains of the electromagnet

4 Visualization and analysis of results

4.1 Hyperparameters

The visualization parameters used in the post-processing stage are summarized in this section. Streamline visualizations are generated with a resolution of 1000 seed points. The maximum streamline length is set to 205, while the maximum number of integration steps is set to 300. A tube filter is applied to the streamlines in order to improve their visibility and readability.

All animations are generated at a frame rate of 25 frames per second. When possible, colormaps are chosen according to semantically. For example, the *Inferno* colormap is used for temperature fields to emphasize high-temperature regions, while other colormaps are selected to clearly represent electrical quantities such as current density, electric field, and electric potential.

4.2 Temperature Field Extraction and Heat Dissipation

The temperature field is visualized using volumetric rendering in order to analyze the thermal distribution inside the device without being limited to surface representations. This approach highlights that heat is distributed within the volume and not only on the external surfaces.

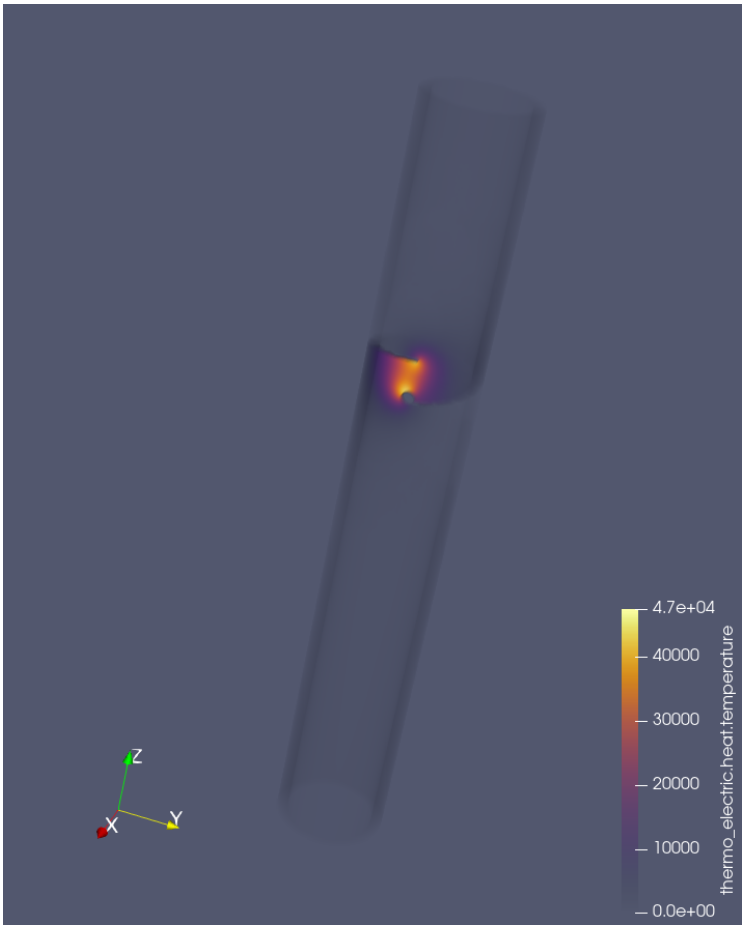


Figure 12: Volumetric visualization of the temperature field highlighting regions of elevated temperature inside the electromagnet.

Figure 12 shows a strong localization of high temperature values at the junction between the helices and the connection ring. This observation indicates that these regions are more thermally loaded compared to the surrounding parts of the device.

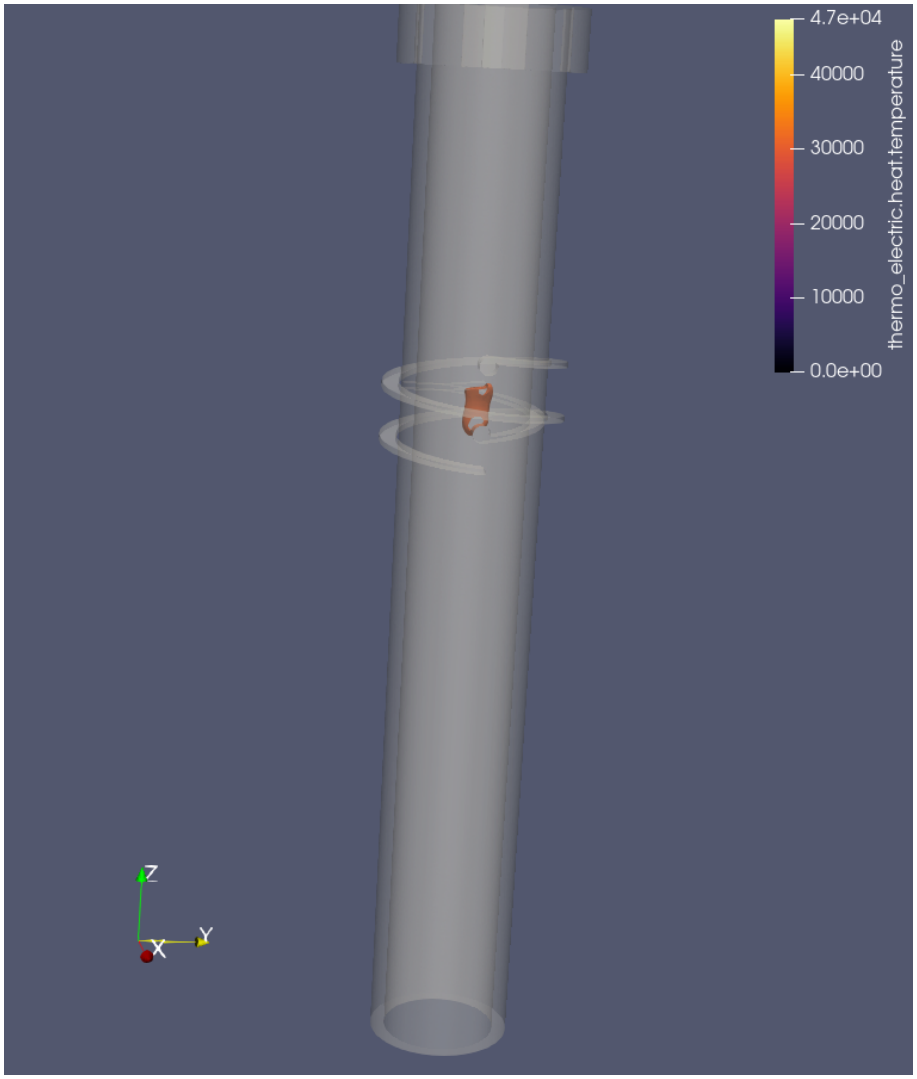


Figure 13: A frame from the animation of the volumetric temperature field illustrating the spatial redistribution of heat over time.

The temperature field is also visualized using an animation (Figure 13), which allows for the spatial redistribution of the temperature to be observed over time. As illustrated by the animation, the regions highlighted in Figure 12 correspond to areas that remain at higher temperature for a longer duration compared to the rest of the domain. The animation shows a progressive spreading of heat within the volume, while elevated temperature levels persist near the helix–ring junction.

4.3 Current Density Distribution

The current density magnitude is visualized using a volumetric color map in order to identify the main electrical conduction paths within the electromagnet.

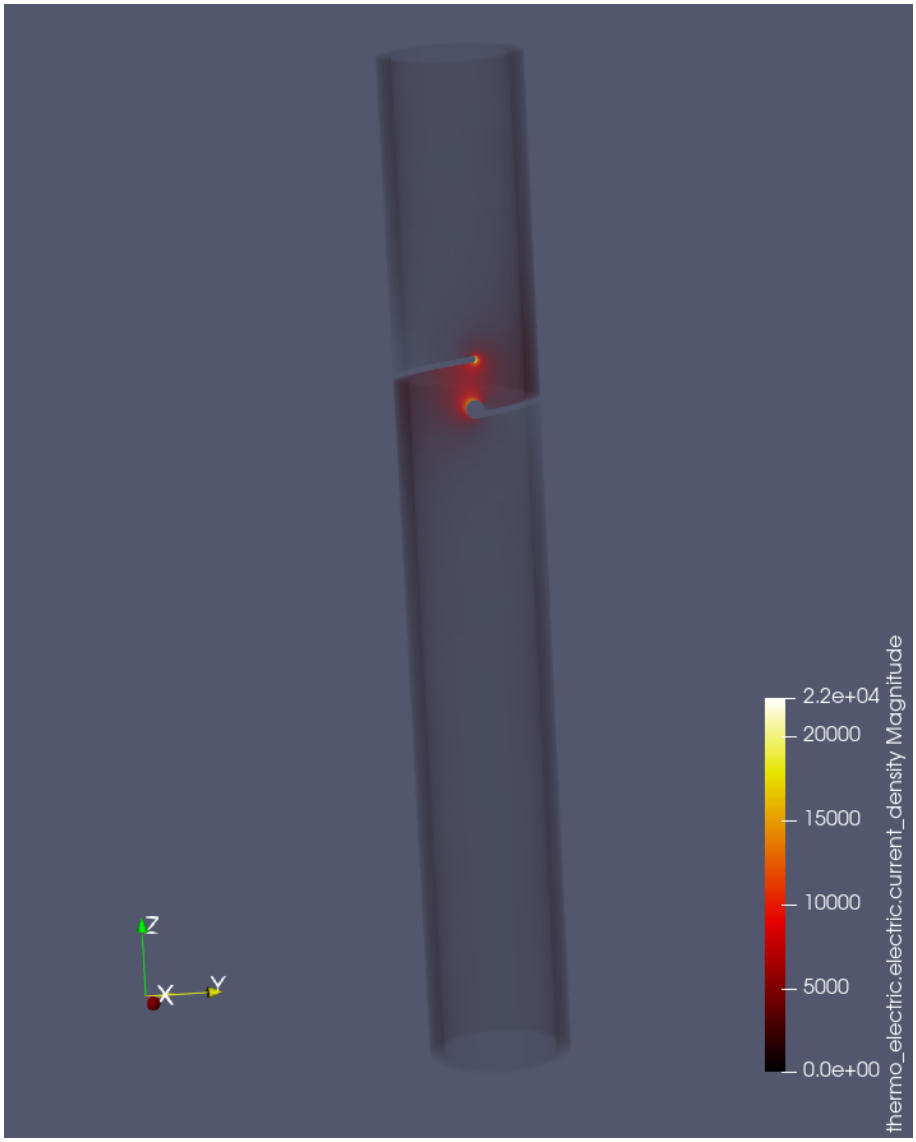


Figure 14: Volumetric visualization of the current density magnitude highlighting regions of high electrical conduction.

Figure 14 shows that the current density is strongly concentrated near the junction between the helices and the connection ring. These regions correspond to areas where the electrical path is constrained by the geometry, leading to locally increased current density. This observation is consistent with the regions of elevated temperature identified previously.

In addition, streamline visualizations colored by current density are used to illustrate the direction of current flow within the device. These streamlines, in figure 15, show that the current follows the geometry of the helices and is strongly deflected near the helix–ring junction, further highlighting the role of geometry in concentrating electrical activity in this region.

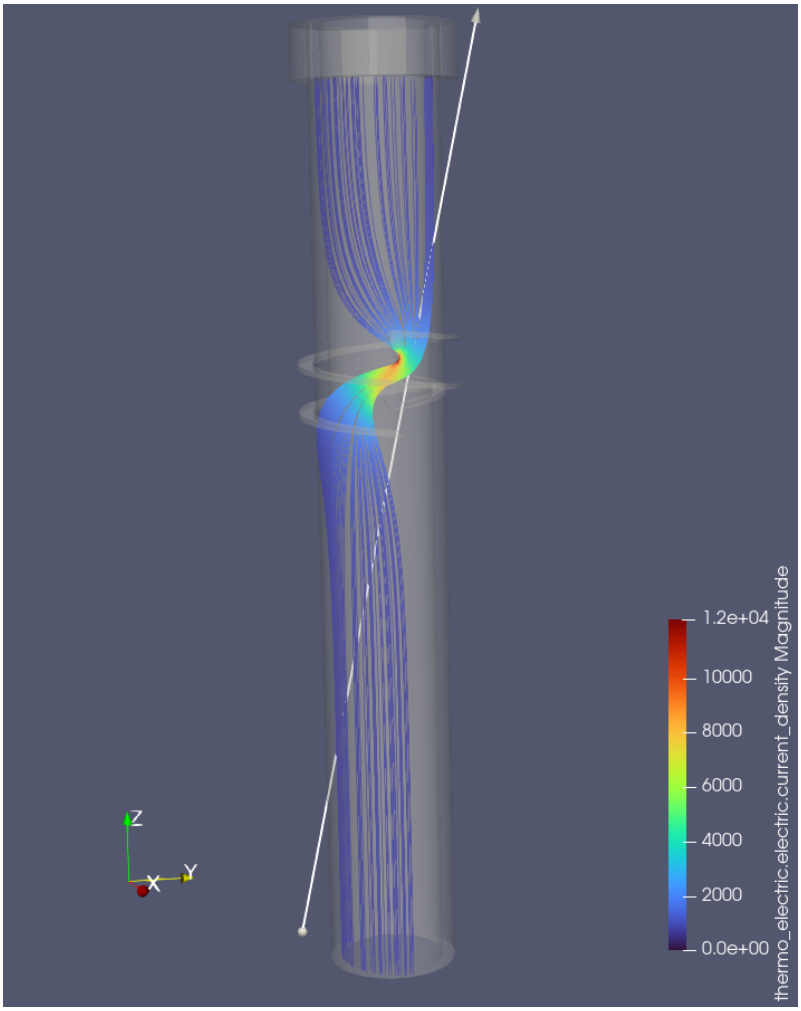


Figure 15: Streamline visualization colored by current density magnitude, illustrating the direction of current flow and its concentration near the helix–ring junction.

4.4 Electric Field Visualization

The electric field is visualized using streamlines colored by the electric field magnitude. This representation allows both the direction and intensity of the electric field to be analyzed within the device.

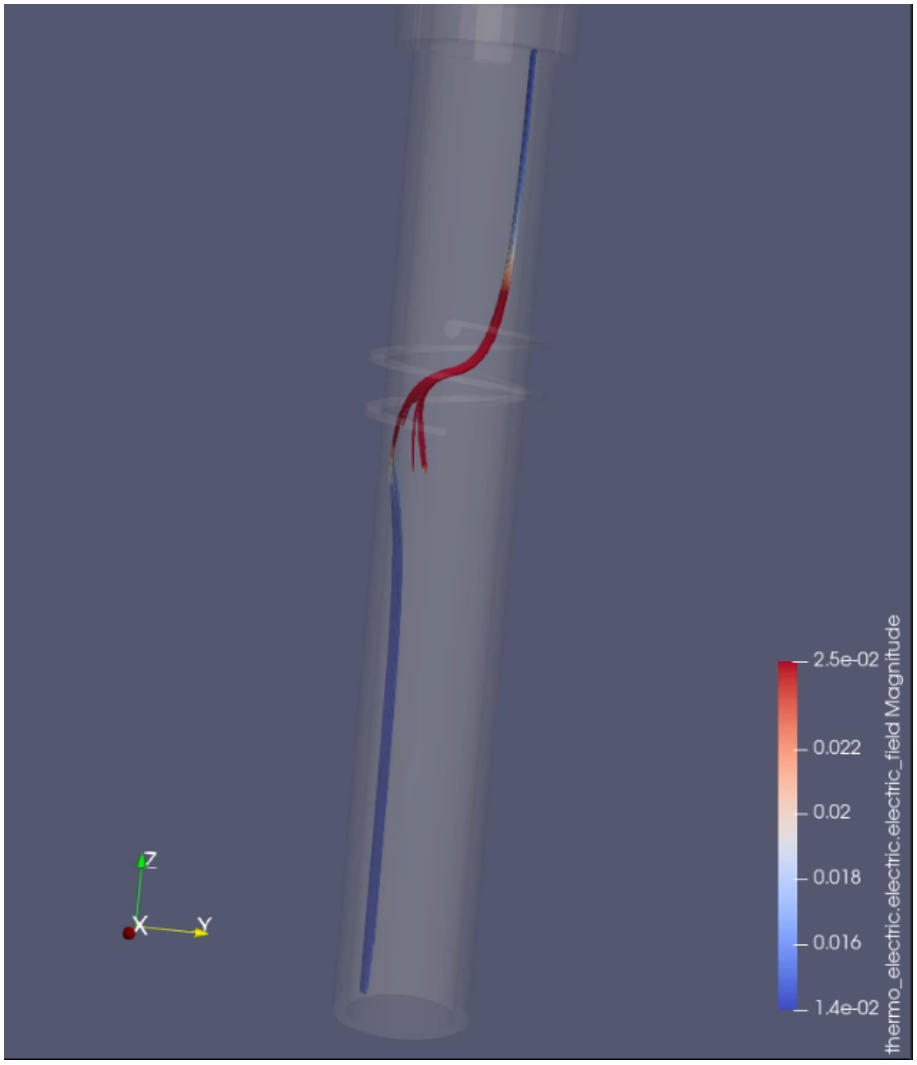


Figure 16: Electric field streamlines colored by field magnitude, illustrating the spatial distribution of the electric field.

Figure 16 shows that the electric field follows the geometry of the conducting paths and intensifies near curved regions and junctions. Higher electric field magnitudes are observed in the helices and near the helix–ring interface, indicating strong potential gradients in these areas. These regions coincide with zones of high current density and temperature, highlighting the coupling between electrical and thermal phenomena.

4.5 Electric Potential Visualization

The electric potential is analyzed using a volumetric animation in order to observe its spatial distribution within the electromagnet over time.

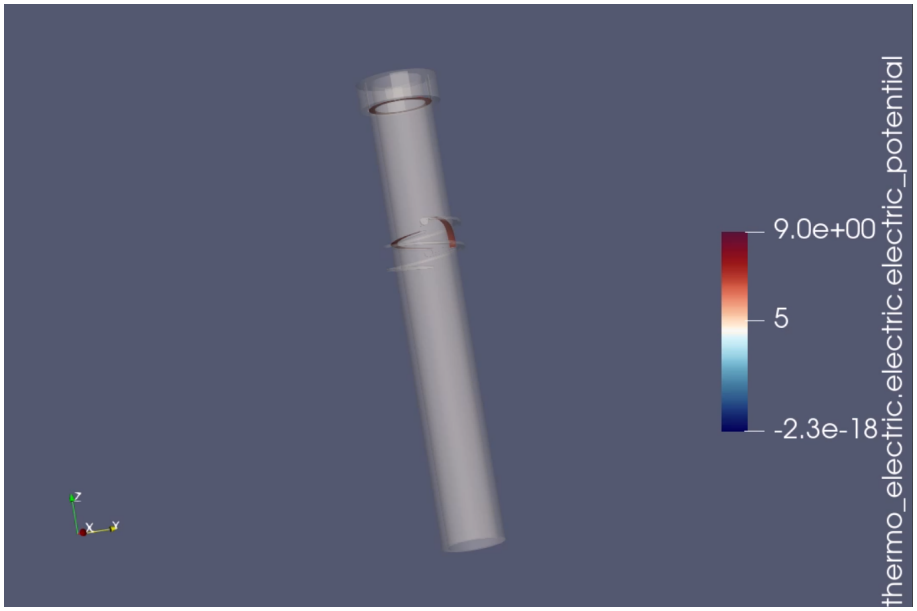


Figure 17: Frame extracted from the animation of the volumetric electric potential distribution inside the electromagnet.

Figure 17 presents a representative frame extracted from the electric potential animation. It illustrates the spatial distribution of the potential at a given time step, with visible variations along the conducting paths, particularly in the helical regions.

The use of animation allows the continuity and evolution of the electric potential to be observed throughout the simulation. The animation shows that the electric potential is lower near the bottom of the electromagnet and higher toward the top, indicating a global potential gradient across the device. Larger potential variations are observed along the helices, particularly near the helix–ring junction. These qualitative observations correlate with the regions of higher current density and elevated temperature discussed in the previous sections.

4.6 Comparison Plots

In order to quantitatively compare the electrical and thermal behaviors of the electromagnet, comparison plots are generated by extracting one-dimensional profiles along two probe lines placed at different locations of the device.

The first probe line is positioned along the connection ring, from the bottom to the top of the electromagnet. This probe allows the global evolution of temperature and current density to be analyzed along the ring.

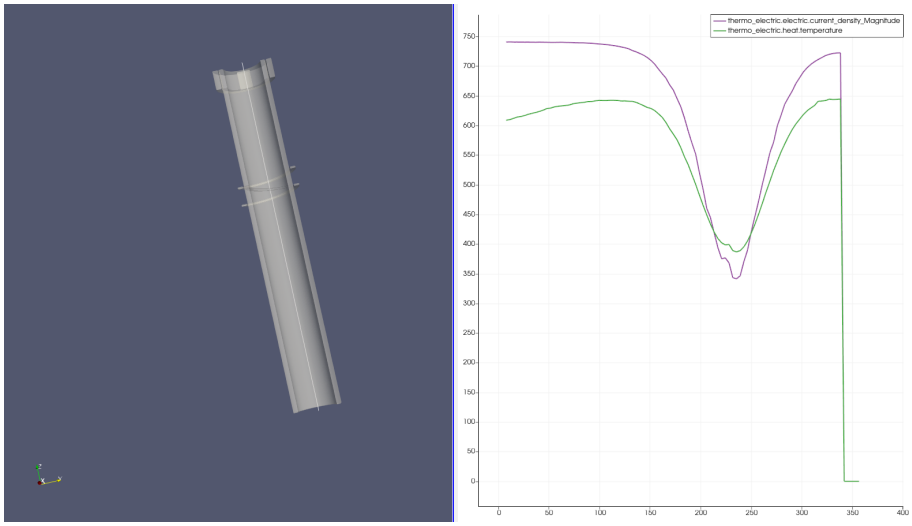


Figure 18: Comparison plot of temperature and current density magnitude extracted along a probe line placed along the connection ring.

Figure 18 shows that temperature and current density exhibit similar trends along the probe line. Both quantities decrease in some regions and increase together in others, indicating a correlation between electrical conduction and thermal behavior, even outside the most thermally active zones.

The second probe line is positioned near the region where elevated temperature values were previously observed, close to the helix–ring junction. This probe is used to further investigate this localized area and to confirm the observations made from the first probe line.

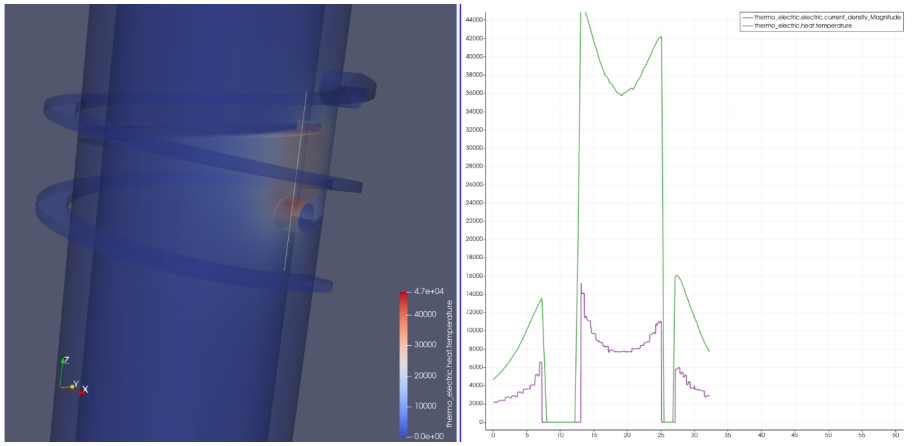


Figure 19: Comparison plot of temperature and current density magnitude extracted along a probe line crossing the helix–ring junction.

Figure 19 shows more pronounced variations of both temperature and current density. Peaks in current density coincide with higher temperature values near the helix–ring junction, confirming that this region is both electrically and thermally active. This focused analysis reinforces the correlation observed in the first comparison plot.

4.7 General Interpretation and Hypothesis

Based on the combined visualizations and comparison plots presented above, a consistent qualitative interpretation of the thermoelectric behavior of the electromagnet can be proposed. The helices, and more specifically the helix–ring junction, systematically appear as regions of elevated

temperature, high current density, strong electric field intensity, and significant variations of electric potential.

The comparison plots indicate that temperature and current density evolve in a correlated manner along both probe lines, with simultaneous increases and decreases observed in the same regions. In particular, the second probe line confirms that this correlation is more pronounced near the helix–ring junction. In addition, the electric potential animation shows larger potential variations along the helices, while these regions still carry a high current density.

These observations suggest that the helical regions, especially near the helix–ring junction, constitute electrically constraining and energetically dissipative paths within the device. The combination of sustained current flow and significant potential variations is consistent with increased electrical losses, which may explain the localized heat accumulation observed in the temperature field.

This interpretation remains qualitative and is based solely on the observed distributions of temperature, current density, electric field, and electric potential. Nevertheless, the consistency between all post-processing results supports the hypothesis that the geometry of the helices plays a dominant role in governing both the electrical and thermal behavior of the electromagnet.

5 Annexes

The animations are provided together with the PDF report. All ParaView visualization states corresponding to the figures presented in this report are also included, enabling reproduction and further inspection of the post-processing results.